There is one shortcoming in the above Huygens' construction for propagation of a wavefront. It could not explain the absence of backward wave which also arises in the above construction. According to electromagnetic wave theory, the backward wave is ruled out inherently. However, Huygens' principle is a good diagrammatic construction which explains the propagation of the wavefront.

7.2.3 Proof for laws of reflection using Huygens' Principle

Let us consider a parallel beam of light is incident on a reflecting plane surface such as a plane mirror XY as shown in Figure 7.4. The incident wavefront is AB and the reflected wavefront is $A'B'$. These wavefronts are perpendicular to the incident rays L, M and reflected rays L', M' respectively. By the time point A of the incident wavefront touches the reflecting surface, the point B is yet to travel a distance BB' to touch the reflecting surface at B'. When the point B touches the reflecting surface at B', the point A would have reached A'. This is applicable to all the points on the wavefront. Thus, the reflected wavefront $A'B'$ emanates as a plane wavefront. The two normals N and N' are considered at the points where the rays L and M fall on the reflecting surface. As reflection happens in the same medium, the speed of light is same before and after the reflection. The time taken for the light to travel from B to B' and A to A' are the same. Thus, the distance BB' is equal to the distance AA'; $(AA' = BB')$.

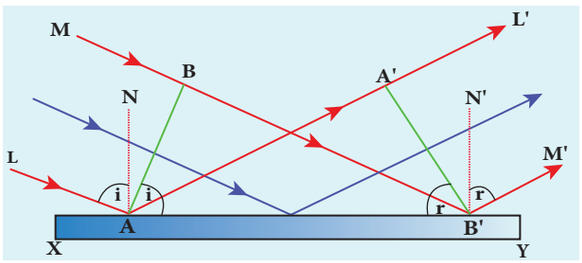


Figure 7.4 Laws of reflection

(i) The incident rays, the reflected rays and the normal are in the same plane.

(ii) Angle of incidence,

$$\angle i = \angle NAL = 90^\circ - \angle NAB = \angle BAB'$$

Angle of reflection,

$$\angle r = \angle N'B'M' = 90^\circ - \angle N'B'A' = \angle A'B'A$$

For the two right angle triangles, $\Delta ABB'$ and $\Delta B'A'A$, the two right angles, $\angle B$ and $\angle A'$ are equal, ($\angle B$ and $\angle A' = 90^\circ$); the two sides, AA' and BB' are equal, $(AA' = BB')$; the side AB' is common. Thus, the two triangles are congruent. As per the property of congruency, the two angles, $\angle BAB'$ and $\angle A'B'A$ must also be equal.

$$i = r \qquad (7.2)$$

Hence, the laws of reflection are proved.

7.2.4 Proof for laws of refraction using Huygens' Principle

Let us consider a parallel beam of light is incident on a refracting plane surface XY such as a glass as shown in Figure 7.5. The incident wavefront AB is in rarer medium (1) and the refracted wavefront $A'B'$ is in denser medium (2). These wavefronts are perpendicular to the incident rays L, M and refracted rays L', M' respectively. By the time the point A of the incident wavefront touches the refracting surface, the point B is yet to travel a distance BB' to touch the refracting surface at B'. When the point B touches the refracting surface at B', the point A would have reached A' in the other medium. This is applicable

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